

En422 : Gas Turbine Engine

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Design of a Supersonic Aerospike Nozzle

PROJECT REPORT

The present report covers our Turbomachinery project tackling an aerospike nozzle theoretical and numerical modelling. From the theoretical output quantities, our objective will be to dissect and model the resulting expansion fan, its reflection, the evolution of the quantities downstream and interpret these results in the global framework of an aerospike nozzle design.

This project allowed us to dive deeper in the last topic of our lecture and learn the fundamental application of the Riemann invariants along the resulting characteristics.

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INTRODUCTION

The purpose of this project is to be able to design and analyze the operation of a supersonic nozzle. In fact, this type of nozzle is common in many industrial fields. We can find such nozzles in civil and military aviation. What is interesting for us is that we also find those nozzles in spacecraft propulsion systems. Indeed, for this applicative case, our group chose to study the theoretical case of the aerospoke nozzle. Like every supersonic nozzle, the operating principle remains the same. It relies on accelerating the flow exiting the combustion or reheating chamber through an isentropic expansion. The velocity difference between the inlet and the outlet is what allows the nozzle to create the thrust needed.

The aerospoke nozzle has not been produced yet and still remains a theoretical solution for spacecraft propulsion, or at best some planar experimentation. Nevertheless, this design proposes performances with way higher efficiencies no matter how the atmospheric pressure changes. This is why we turned our interest into this subject instead of more common nozzle geometry. But this choice brings some variations in what we are used to work with. Actually, the throat marks the end of the closed part of the nozzle. The flow undergoes a violent variation in the geometry and since it becomes supersonic at this very point, expansion fans are created from there. Those expansion fans then propagate in contact with the outside air until they reach continuous equilibrium with the current atmospheric pressure. This implies that a new expansion fan appears for each new altitude. Knowing this phenomenon, we will aim to determine the initial expansion fan as well as its interaction with the wall of the nozzle. For our study, we will first consider that the wall is a straight line with a constant angle of β_{init} . Then, we will work on a more complex geometry where the wall is a curved spline adapted depending on the expansion fan to keep the fluid tangent in any moment and optimize thrust and limit losses.

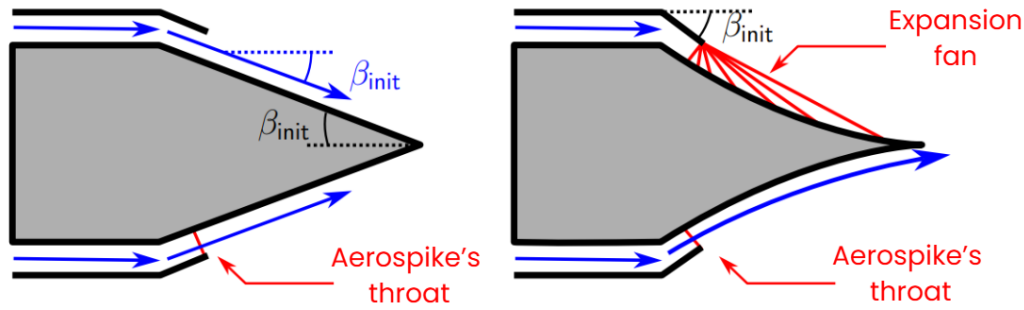
Since a three-dimensional work would take too much time and will be too complex for us to analyze properly, we will place ourselves in a two-dimensional planar supersonic framework.

All the data for our computation are derived from our latest tutorial, especially the second part focusing on nozzles. To summarize the values used in this project, we can refer to the following table.

$\dot{m}_{air}$ [kg.s ⁻¹]	M_{∞}	$P_{s,\infty}$ [kPa]	$P_{t,6}$ [kPa]	$T_{t,6}$ [K]	H/w
102	0.0	26.5	628	2170	0.5

I – THERMODYNAMIC STUDY

The objective of this section is to compute the first thermodynamic properties of the fluid especially when considering the nozzle adapted. We want to end up knowing the gross thrust of the aerospike nozzle. For that, we will go through different computation steps. The different geometries can be found in this figure:



A – REAL NOZZLE

To begin with, we want to know what the throat area is in order to know the throat height H_{th} . For that, we consider that the nozzle is not cylindrical but linear symmetrical as in the X-33 project of Lockheed Martin in 1999.

Using the mass flow parameter formulation, we are able to isolate the throat's surface A_{th} :

$$MFP(M_{th}) = \frac{\dot{m}_{air} \sqrt{T_{t,6}}}{P_{t,6} \cdot A_{th}} = \sqrt{\frac{\gamma}{r}} \cdot M_{th} \cdot \left(1 + \frac{\gamma-1}{2} M_{th}^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

$$\Leftrightarrow A_{th} = \frac{\dot{m}_{air} \sqrt{T_{t,6}}}{P_{t,6}} \times \sqrt{\frac{r}{\gamma}} \cdot \frac{1}{M_{th}} \cdot \left(1 + \frac{\gamma-1}{2} M_{th}^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Knowing that at the throat, the Mach number is 1, and considering $\gamma = 1.4$, we finally have:

$$A_{th} = 0.187 \text{ m}^2$$

Since we chose to work with a linear nozzle, the surface is only the product of the height by the length of the throat. We also have the aspect ratio given in the data. Therefore:

$$\begin{cases} A_{th} = 0.187 \\ \frac{H_{th}}{w} = 0.5 \end{cases} \Leftrightarrow \begin{cases} H_{th} = 0.3058 \text{ m} \\ w = 0.6115 \text{ m} \end{cases}$$

We finally have a real height of $H_{th} = 0.3058 \text{ m}$

B – PROPERLY ADAPTED NOZZLE

From now on, we consider our nozzle to be adapted as explained at the beginning of the thermodynamical study. We now want to compute the Mach number at the exit of the nozzle. This Mach number corresponds to the one after passing through the expansion fan. It is the one region where pressure connects to the atmospheric value. We will consider station 6 as the one after passing the expansion fan.

The total quantities are well known, having no heat transfer or moving parts in our nozzle gives the conservation of those quantities all along the nozzle. The nozzle being properly adapted means that our static pressure at station 6 is equal to the atmospheric pressure $P_{s,\infty}$.

Using the Saint-Venant equation, we can isolate the Mach number for the adapted nozzle M_{adapt} :

$$P_{t6} = P_{atm} \left(1 + \frac{\gamma - 1}{2} M_{adapt}^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

$$\Leftrightarrow M_{adapt} = \sqrt{\frac{2}{\gamma - 1} \left[\left(\frac{P_{t6}}{P_{atm}} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right]}$$

We end up with a Mach number of: $M_{adapt} = 2.71$

With this new Mach number, we can compute the theoretical area we should have, again using the mass flow parameter formulation:

$$A_{theo} = \frac{\dot{m}_{air} \sqrt{T_{t,6}}}{P_{t,6}} \times \sqrt{\frac{r}{\gamma}} \cdot \frac{1}{M_{adapt}} \cdot \left(1 + \frac{\gamma - 1}{2} M_{adapt}^2 \right)^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

After the numerical application, we obtain a theoretical area of: $A_{theo} = 0.60 \text{ m}^2$

As explained above in the introduction, the nozzle produces thrust by accelerating air flow. The higher the velocity difference between the intake and the outlet, the higher the thrust is. Since the objective of this part is to compute the gross thrust of the nozzle, we need to compute the exhaust velocity. To do so, we know that the mass flow rate is conserved throughout the nozzle, and we just computed the theoretical area of the exit so:

$$\dot{m}_{air} = \rho v_{ex} A_{theo}$$

$$\Leftrightarrow v_{ex} = \frac{\dot{m}_{air}}{\rho A_{theo}} = \frac{\dot{m}_{air} r T_{s,6}}{P_{s,\infty} A_{theo}}$$

Note that static pressure is still equal to the atmospheric one because of the adapted nozzle assumption. Also, the static temperature can be computed using the Saint-Venant equation and gives us:

$$T_{s,6} = T_{t,6} \left(1 + \frac{\gamma - 1}{2} M_{adapt}^2 \right)^{-1} = 878.96 \text{ K}$$

We finally obtain an exhaust velocity of:

$$v_{ex} = 1618.28 \text{ m.s}^{-1}$$

C – GROSS THRUST

We now have all the variables we need to compute the gross thrust of our adapted aerospike nozzle. Since it is adapted, the pressure difference term in the formulation of the thrust is equal to zero and only the velocity difference matters:

$$T_g = \dot{m}_{air} \cdot (v_{ex} - v_{in})$$

In our case, we consider the nozzle fixed at sea-level. Therefore, the inlet velocity is neglected and we find:

$$T_g = \dot{m}_{air} \cdot v_{ex} = 165.1 \text{ kN}$$

Note that we do not consider any losses in the nozzle, which is not the real case obviously. We also saw that the real section was way smaller than the ideal one, which makes the final real thrust smaller.

II – AERODYNAMIC DESIGN

At the throat, we have a direct change in the geometry of the wall guiding the flow. It is precisely this change that generates Prandtl-Mayer expansion fan. When the flow passes through this expansion fan, it is continuously turned and accelerated until it reaches the final state like explained earlier in this report.

Therefore, this second part will be dedicated to a deeper study of the expansion fans. We will try to understand their formation from the throat and how the characteristics reflect when they encounter the rigid wall. Our goal is to discretize the expansion fan at the throat in a finite number of characteristics n and to compute the properties in each region created by the intersection of falling and rising characteristics. To do so we will mainly use the Prandtl-Mayer function:

$$v(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \cdot \tan^{-1} \left(\sqrt{\frac{\gamma-1}{\gamma+1}} (M^2 - 1) \right) - \tan^{-1} (\sqrt{M^2 - 1})$$

A – FLOW DEFLECTION ANGLE

The flow deflection angle is the value of the Prandtl-Mayer function for the Mach number after the expansion fan, which in our case, corresponds to M_{adapt} . Using the Prandtl-Mayer function:

$$v(M_{adapt}) = v_{adapt} = 0.765 \text{ rad} = 43.83^\circ$$

Even if this value is expressed in angle, it is quite difficult to visualize it, and we therefore only take it as result of the Prandtl-Mayer function and it will not be represented in our different schemes.

B – MACH ANGLE

We know that the ejector is oriented toward the central body at a certain angle labeled β_{init} . First, let us deduce this angle that will impact all the following computation of the regions. We know:

$$\beta_{init} = \frac{v_{adapt}}{2} = \frac{43.83}{2} = 21.92^\circ$$

Still considering the Mach number downstream of the expansion fan to be our M_{adapt} , we can also compute our Mach angle μ_{adapt} which represents the angle between the velocity vector of the flow and the characteristic considered straight in our model.

$$\mu_{adapt} = \sin^{-1} \left(\frac{1}{M_{adapt}} \right) = \sin^{-1} \left(\frac{1}{2.71} \right) = 21.65^\circ$$

This value is what the flow sees in its own referential. We can also determine the Mach angle in the absolute referential, comparing it to the horizontal axis. We then have:

$$\mu_{adapt,abs} = \beta_{init} - \mu_{adapt} = 0.3^\circ$$

As we can see, the flow almost comes out horizontally after the expansion fan.

C – CHARACTERISTICS REFLECTION

The reflection of a characteristic relies on the impermeability of the boundary. Indeed, the flow can't penetrate the wall, so the velocity must always remain tangent to it. The mechanism can be detailed more precisely. A falling characteristic carries a Riemann invariant, which contains 'instructions' about changes in flow direction. If the wall is a straight ramp and follows the 'instruction', it would lead to two physically impossible scenarios: penetration of the wall or flow separation.

To satisfy the tangency condition, a rising characteristic is generated at the reflection point. It carries an equal and opposite flow direction, ensuring that the flow remains tangent to the wall. The superposition of the incident and reflected waves ensures that the resultant flow angle matches the local wall angle.

D – DISCRETIZATION OF THE INITIAL EXPANSION FAN

In the infinite number of characteristics composing the expansion fan occurring at the upper lip, we can approach the modelling of this physics by discretizing the fan with n characteristics. Reflecting on the nozzle surface, the falling and rising characteristics will create a multi-region network transporting varying thermodynamic quantities: M , T_s , P_s , etc... To determine the evolution of these quantities, let us first establish the theory behind this downstream network.

$$\begin{cases} i_+ \rightarrow \text{rising} \\ i_- \rightarrow \text{falling} \end{cases}$$

Let us consider that a characteristic is labeled with the index i . The next characteristic is then called j with the relation $j = i+1$ with the sign convention:

To count the number of regions created for n characteristics, we see that a pattern repeats. Each rising characteristic i_+ crosses the falling $i_- + 1$, $i_- + 2$, $i_- + 3$. In the end, all the crossed falling characteristic indices belong to the interval: $[[i + 1 ; n]]$. This operation generates n^2 regions. Besides those created regions, we also need to consider all the regions created at the wall of the nozzle. For that, we know that each characteristic i_- will create one region bordered by the wall, the j_- and the i_+ . In the end, with n characteristics, we need to add n more regions to the initial counting. Lastly, since we are bordered by a wall, we only see half of the possible regions created so for our final counting, we need to divide the sum of the two previous results by two. Our final formulation for the number of regions gives:

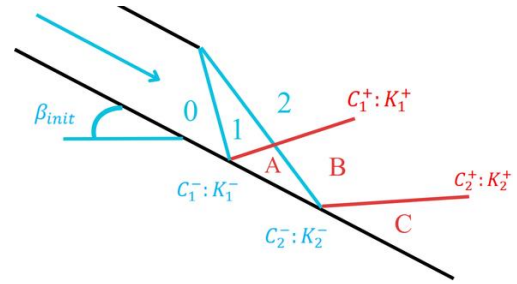
$$N_{regions} = \frac{n^2 + n}{2}$$

Note that we validated this formulation in our program by summing the number of regions created by the algorithm and simultaneously computing the result of our theoretical formulation: both are equal.

Now that the general characteristic intersection network framework is in place, the fundamental step to implementing it algorithmically is to perform an analytic resolution for a simple example system, here composed of 2 characteristics: **blue** and **purple** variables respectively represent **known** and **unknown** quantities.

i. Falling characteristics & Falling gradient regions

These first characteristics properties are known to us, since we decide how we discretize the characteristics leading to the final flow state. For our study, we will discretize the flow deflection angle, with ν_i therefore ranging from ν_{init} to ν_{adapt} . The resulting M_i and μ_i quantities are direct, by numerically solving the Prandtl-Mayer function and the arcsine of the inverse of the Mach. Note that we also store these quantities for the later display of the network and regions created.



In our $i \in [1,2]$ characteristics example, the Riemann invariants transported by each characteristic therefore gives us the following equations.

$$K_1^- = \beta_1 + \nu_1 = (\beta_{init} + \nu_1) + \nu_1 = \beta_{init} + 2\nu_1 \quad \text{and same for } K_2^-$$

These falling characteristic, upstream of the first reflection and intersection with the rising characteristics, create a gradient of “falling Mach” within the first regions (regions 1 and 2 in our example). As we know the quantities defining these regions (flow deflection and Mach cone angles), the thermodynamic quantities can also be known with T and P solved by inverting the previously presented St-Venant equations.

Numerically, these quantities are stored in a `reg_m` dictionary as such:

```
for i in range(nb_char):
    reg_m[i]={"Mach_i":Machs_m[i],"P_i":press(Machs_m[i]),"T_i":temp(Machs_m[i])}
```

ii. Reflecting characteristics & Nozzle boundary regions

At the nozzle surface, the Riemann invariant reads (with A for $i=1$):

$$K_1^- = \beta_A + \nu_A = \beta_{init} + \nu_A \quad \text{allowing us to establish } \boxed{\nu_A = 2\nu_i}$$

Similarly, now that we solved the rising deflection angle, we can solve for M_A and μ_A and store these quantities for the later display of the network and regions.

The reflection points coordinates can be calculated geometrically, by computing the slope of the falling characteristic and by knowing the fan coordinates:

$$slope_{C_i^-} = \frac{y_r - y_{fan}}{x_r - x_{fan}} = \frac{h \cdot slope_{nozz}}{x_r - x_{fan}} \Leftrightarrow \boxed{x_r = x_{fan} + \frac{h \cdot slope_{nozz}}{slope_{C_i^-}}} \quad \boxed{y_r = x_r \cdot slope_{nozz}}$$

At these coordinates, new boundary regions are created between the rising characteristics and the nozzle surface (regions A and C in our example), downstream of the reflecting point. Thanks to the two previous equations, we now know these regions' coordinates and thermodynamic quantities.

We also store these values in a `reg_nozz` dictionary as such:

```
for i in range(nb_char):
    # Calculation of x_r and y_r are not displayed here
    reg_nozz[i]={"Mach_i":Machs_p[i],"P_i":press(Machs_p[i]),"T_i":temp(Machs_p[i]),
    "xy_r":{x_r,y_r}}
```

iii. Intersecting characteristics & Quadrilateral network regions

Recalling that the rising characteristic Riemann constant reads $K_i^+ = \beta_i - v_i$, the resulting intersection of C_i^- and C_j^+ for the (i,j) region gives us the 2-equations 2-unknowns system:

$$\begin{cases} K_i^- = \beta_{init} + v_i = \beta_{ij} + v_{ij} \\ K_j^+ = \beta_{init} - v_j = \beta_{ij} - v_{ij} \end{cases} \Rightarrow \boxed{\begin{aligned} \beta_{ij} &= \beta_{init} + v_i - v_j \\ v_{ij} &= v_i + v_j \end{aligned}}$$

Once again, it is now possible to solve for the regions Mach and Mach cone angle M_{ij} and μ_{ij} . Furthermore, the inverted St-Venant equations allow us to solve for the static Pressure and Temperature inside these regions. Lastly, we can establish the coordinates of these intersection points geometrically, by equaling the rising i^+ and falling j^- characteristic line equations:

$$\begin{cases} y_{C_i^+} = y_r + slope_{C_i^+}(x - x_r) \\ y_{C_j^-} = y_{fan} + slope_{C_j^-}(x - x_{fan}) \end{cases} \Rightarrow \boxed{\begin{aligned} x_{int} &= \frac{y_{fan} - y_r + slope_{C_i^+} \cdot x_r}{slope_{C_i^+} - slope_{C_j^-}} \\ y_{int} &= y_{fan} + slope_{C_j^-} \cdot (x_{int} - x_{fan}) \end{aligned}}$$

Numerically, our loop is now defined with a second j -loop for the falling characteristics crossing the $i_+ + 1$, $i_+ + 2$, $i_+ + 3...$ rising characteristics. The quantities defining the (i,j) regions are once again stored in a `reg_int` dictionary as such:

```
for i in range(nb_char):
    for j in range(i+1,nb_char):
        B_ij=B_init+nus[i]-nus[j]
        nu_ij=nus[i]+nus[j]
        # Calculation of x_r, y_r, x_int, y_int are not displayed here
        reg_int[(i,j)]={"B_ij":B_ij,"nu_ij":nu_ij,"Mach_ij":solve_M(nu_ij),"xy_int":
        [x_int,y_int],"P_ij":press(Mach_ij),"T_ij":temp(Mach_ij)}
```

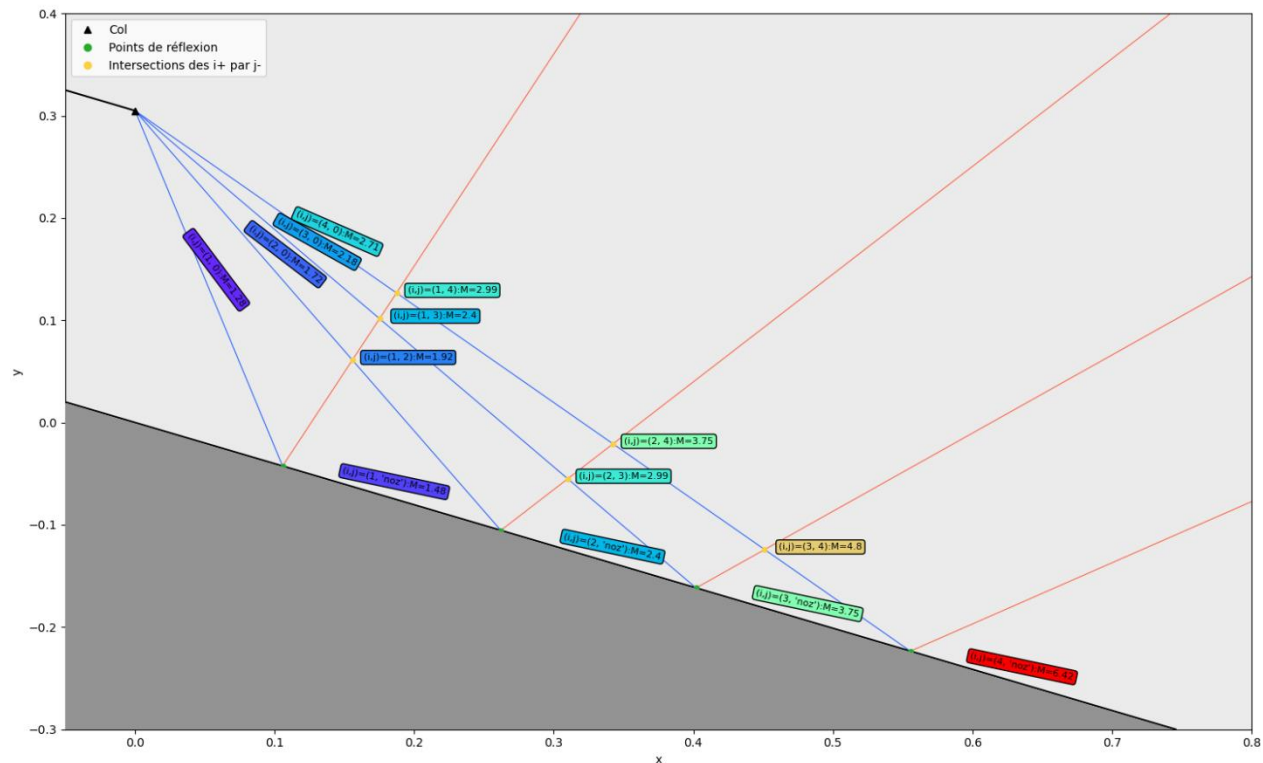
iv. Global expansion fan model

The intermediate display of the quantities shows the following results:

```

Région (i+,j-)=(1, 2): Mach=2.99 / T=778.28K / P=17.35kPa
Région (i-,0)=(1, 0): M=1.28 / T=1632.04K / P=231.69kPa
Région (i-,0)=(2, 0): M=2.71 / T=878.96K / P=26.56kPa
Région (i+,-,nozzle)=(1, 'nozzle'): Mach=1.48 / T=1505.95K / P=174.86kPa
Région (i+,-,nozzle)=(2, 'nozzle'): Mach=6.42 / T=234.86K / P=0.26kPa
  
```

We observe an increasing Mach number downstream of the flow, and a region 1 to 2 Mach number gradient leading to the imposed $M_{adapt} = 2.71$. Now that our model appears coherent, we can enhance the graphical display and increase the number of characteristics to obtain the following Mach number gradient. Furthermore, the near-like gradient for a non-labelled high n_{char} computation as well as static Pressure and static Temperature labelled gradients are in appendices.



Overall, the above and appendices gradients plot correctly capture the expected behavior and thermal – kinetic energy transformation downstream of the upper lip initial expansion fan. With the acceleration of the flow after each reflection, we observe a high 6.42 Mach number in the last downstream region, while the temperature and pressure (following St-Venant relation) are at their lowest in this region.

We can therefore successfully validate the physical coherence of this isentropic expansion numerical model, correctly representing the evolution of this supersonic expansion flow quantities.

E – ADAPTED GEOMETRY

In practice, the geometry of the central body is designed to prevent the characteristics reflection. The geometry is not a linear ramp anymore but represented by a spline. The curvature of the spline is defined at each point on the flow direction parameter. Indeed, the incident flow should be tangent at each point to the central body, physically translating to:

$$\beta_{curve} = \beta_{ij\ local}$$

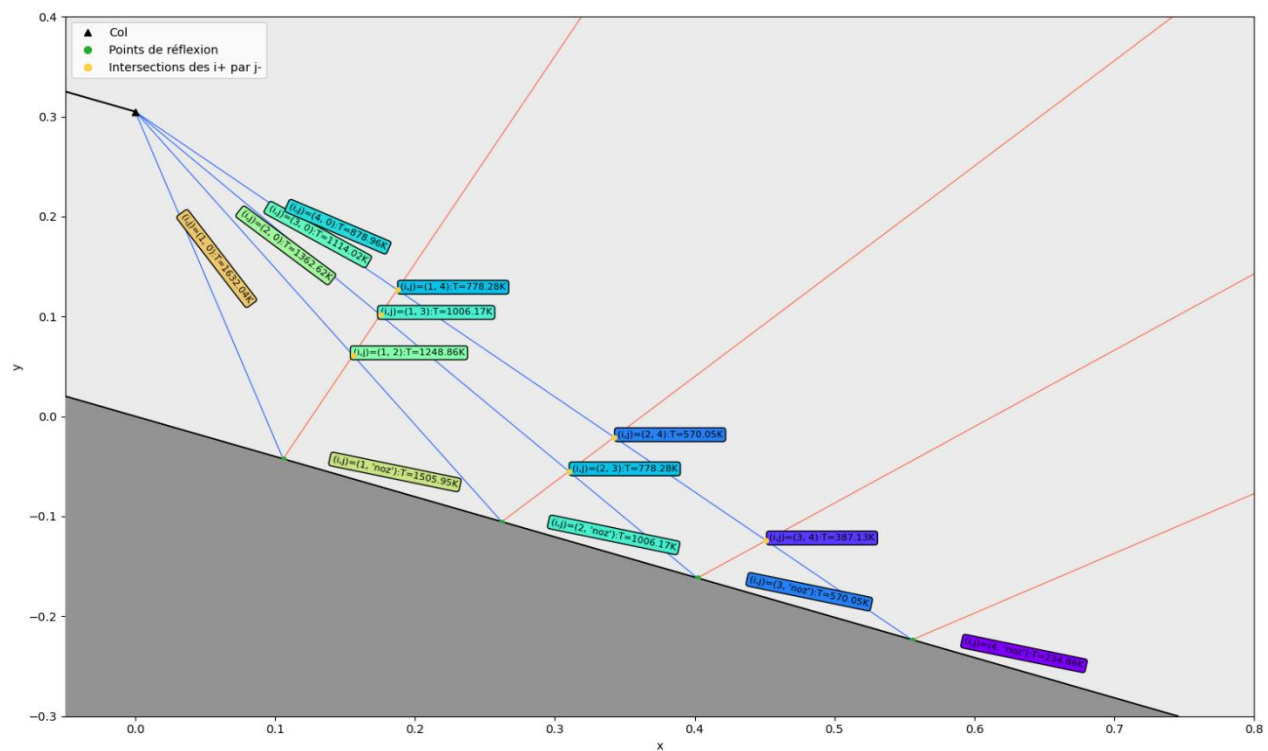
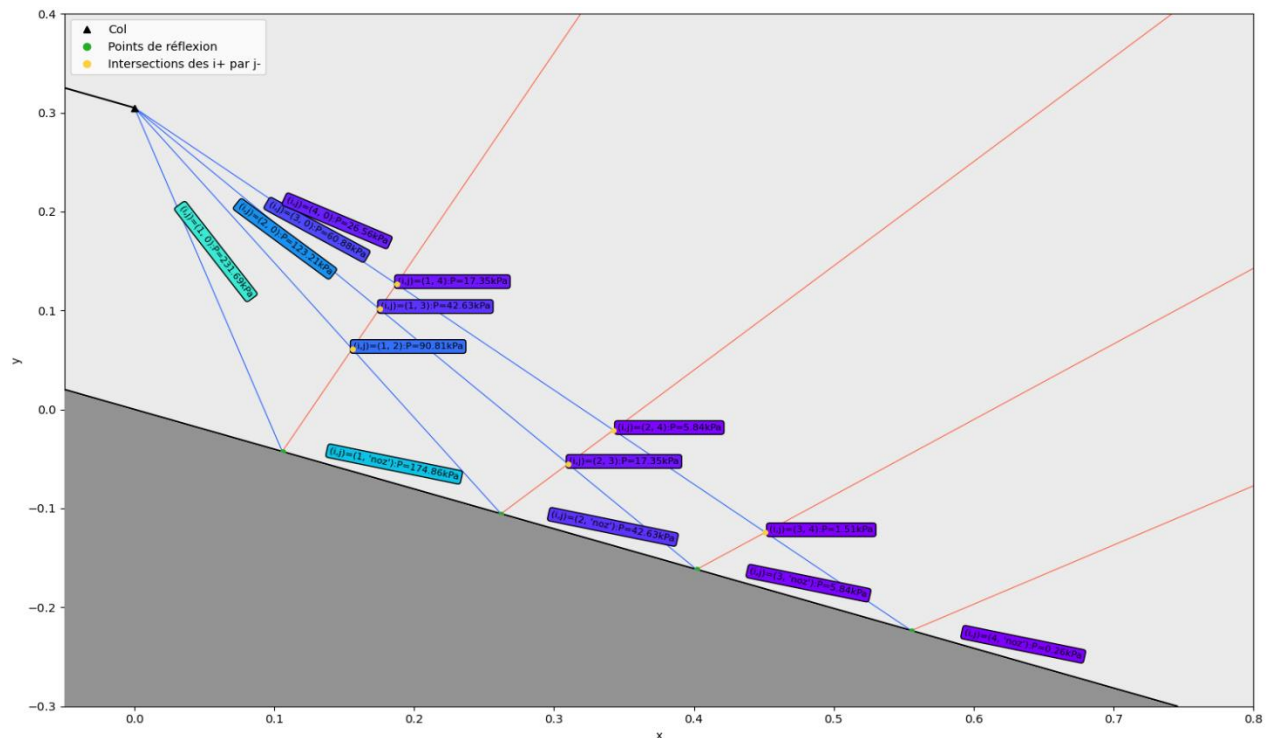
Therefore, here the flow imposes the geometry whereas in the previous section, the geometry imposed the flow. This would allow a higher exhaust velocity, therefore a higher thrust force. Also, we can expect a larger exit section because following the tangent flow, it will move away from the upper wall. This will give a higher Mach number at the exit. If the Mach number is higher, the velocity exit resulting is more important and the thrust increases.

However, with our numerical algorithm based on a fixed β_{init} both for the region quantities calculation AND graphical lines plotting, we did not manage to figure out how we could implement this new geometry iterative modification properly. Therefore, the computation of the thrust (which we expected to do my adding the multiplication of the nozzle regions pressure by the spline surface) was also not possible, requiring the region computation code to be possible.

An ideal solution would have been to restructure the global code logic to make it fundamentally based on this varying $\beta_{ij\ local}$, a solution for which we did not have the time to implement manually nor the will to implement a Gen-AI created result which we would not have been able to understand or explain. We therefore chose to stick with the first model framework method: only handing in results for which we understand the background physics and modelling logic.

Note: It could have also been possible to improve the accuracy of the model by taking into account the fact that the characteristics are in reality curves and not straight lines.

APPENDICES



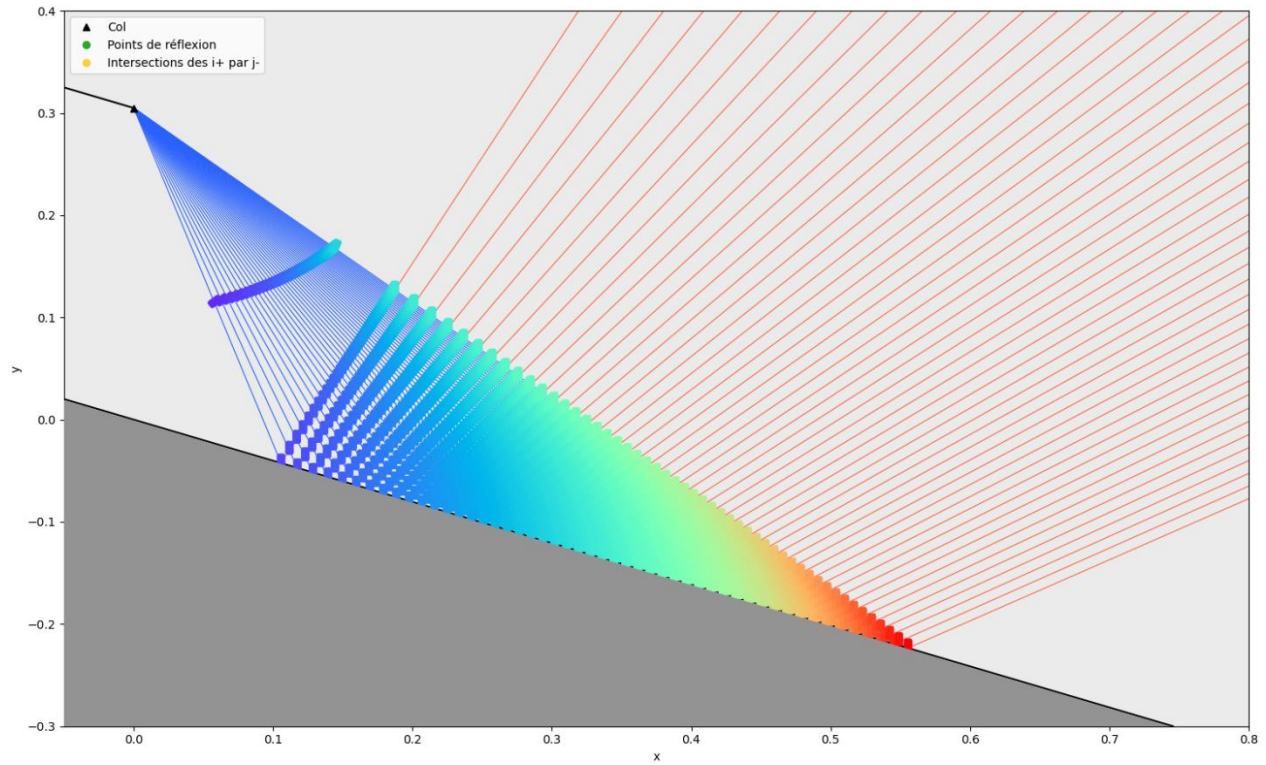


Figure 3 - High n_{char} computation Mach gradient

END OF REPORT